

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 November 2023

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2023 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. Kinetic energy depends on [1]

- A) volume only
- B) mass and speed
- C) colour only
- D) charge only

2. The halogens are in [1]

- A) Group 1
- B) Group 0
- C) Period 1 only
- D) Group 7 / 17

3. The SI unit of speed is [1]

- A) N
- B) kg
- C) J
- D) m/s

4. Salivary amylase begins the digestion of [1]

- A) fat to glycerol
- B) protein to amino acids
- C) fibre to glucose
- D) starch to maltose

5. Neutral solutions have pH [1]

- A) 0
- B) 14
- C) 1
- D) 7

6. The main function of red blood cells is to [1]

- A) clot blood
- B) fight all pathogens by phagocytosis only
- C) make urea
- D) transport oxygen

7. Red blood cells are adapted to carry oxygen because they contain [1]

- A) chlorophyll
- B) starch
- C) cellulose
- D) haemoglobin

8. Aerobic respiration in humans is summarised by [1]

- A) carbon dioxide + water → glucose + oxygen
- B) glucose + oxygen → carbon dioxide + water + energy
- C) protein + oxygen → urea
- D) glucose → lactic acid only

9. A dull black surface, compared with a shiny silver surface, [1]

- A) is a poor absorber of radiation
- B) cannot emit radiation
- C) reflects all heat always
- D) is a good absorber and emitter of radiation

10. A vacuum flask reduces heat loss by [1]

- A) increasing surface area
- B) making the tea denser
- C) electrolysis
- D) convection and conduction (vacuum) and radiation (silvering)

11. A louder sound has a greater [1]

- A) frequency only
- B) speed in the same medium always because it is loud
- C) amplitude
- D) wavelength always

12. Which material is magnetic? [1]

- A) copper
- B) iron
- C) plastic
- D) wood

13. During electrolysis, cations move to the [1]

- A) fuse
- B) cathode
- C) anode
- D) voltmeter

14. An object has mass 20 g and volume 20 cm³. Its density is [1]

- A) 1 g/cm³
- B) 20 g/cm³
- C) 40 g/cm³
- D) 400 g/cm³

15. Protease enzymes digest proteins to [1]

- A) amino acids
- B) fatty acids
- C) maltose
- D) glucose

16. A molecule of carbon dioxide contains [1]

- A) ionic bonds
- B) no bonds
- C) metallic bonds
- D) covalent bonds

17. The lock-and-key model says the substrate fits the [1]

- A) cell wall
- B) vacuole
- C) active site
- D) nucleus

18. The particle with a positive charge in the nucleus is the [1]

- A) molecule
- B) electron
- C) proton
- D) neutron

19. Tuberculosis (TB) is caused by a bacterium and is spread mainly by [1]

- A) ticks
- B) contaminated rivers only
- C) droplet infection (coughing, sneezing)
- D) mosquitoes

20. A covalent bond is [1]

- A) an ionic lattice
- B) the transfer of electrons
- C) a shared pair of electrons
- D) a sea of protons

21. A $4\ \Omega$ resistor is connected to a 6 V supply. The current is [1]

- A) 24 A
- B) 10 A
- C) 4 A
- D) 1 A

22. Platelets are needed for [1]

- A) photosynthesis
- B) making bile
- C) blood clotting
- D) osmoregulation only

23. Which reaction is correct? [1]

- A) acid + metal → salt + water only
- B) acid + metal → salt + hydrogen
- C) acid + carbonate → salt + hydrogen
- D) acid + alkali → salt + oxygen

24. Noble gases (Group 0/18) are [1]

- A) unreactive because they have a full outer shell
- B) alkali metals
- C) very reactive metals
- D) halogens

25. In humans, fertilisation normally occurs in the **[1]**

- A) cervix
- B) uterus wall only
- C) oviduct / Fallopian tube
- D) ovary

26. Alkalis in aqueous solution contain **[1]**

- A) Cl^- only
- B) H^+
- C) OH^-
- D) Na^+ only

27. A force of 50 N acts on an area of 5 m². The pressure is **[1]**

- A) 250 N/m²
- B) 55 N/m²
- C) 10 N/m²
- D) 5 N/m²

28. Each enzyme works best at its **[1]**

- A) optimum pH and temperature
- B) boiling point
- C) ignition temperature
- D) freezing point

29. Which is correct? **[1]**

- A) silver displaces sodium
- B) gold displaces zinc
- C) copper displaces magnesium from $\text{MgSO}_4(\text{aq})$
- D) magnesium displaces copper from $\text{CuSO}_4(\text{aq})$

30. A leaf that has photosynthesised is tested with iodine. A blue-black colour shows **[1]**

- A) starch
- B) protein
- C) reducing sugar
- D) lipid

31. In human muscle during vigorous exercise, anaerobic respiration produces **[1]**

- A) oxygen
- B) ethanol only
- C) carbon dioxide and water
- D) lactic acid

32. Which is a correct order in treating river water for drinking? **[1]**

- A) chlorination → screening
- B) boiling → mining
- C) electrolysis → rusting
- D) screening → filtration → chlorination

33. The SI unit of pressure is the **[1]**

- A) joule
- B) watt
- C) pascal
- D) newton

34. In a food chain, a producer is an organism that **[1]**

- A) eats animals only
- B) makes its own food by photosynthesis
- C) is always a fungus
- D) breaks down dead matter only

35. The SI unit of force is the **[1]**

- A) newton
- B) joule
- C) kilogram
- D) watt

36. Average speed is **[1]**

- A) distance $\times$ time
- B) mass / time
- C) distance / time
- D) time / distance

37. Which statement is correct? **[1]**

- A) Xylem carries blood
- B) Capillaries have thick muscular walls
- C) Veins have the highest blood pressure
- D) Arteries carry blood away from the heart

38. In water treatment, filtration **[1]**

- A) adds fluoride only
- B) boils the water
- C) kills all viruses only
- D) removes solid particles

39. A neutron has **[1]**

- A) charge +2
- B) charge +1
- C) charge -1
- D) no charge

40. Ohm's law may be written **[1]**

- A) $P = I/V$
- B) $V = IR$
- C) $R = VI$
- D) $V = I/R$

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).